

KALP PAREKH

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IMPACT HOOK

SDE with production experience scaling backend systems from 0→50K concurrent users. Built full-stack SaaS platform processing 200K+ monthly transactions with 99.7% uptime. Expertise: Distributed systems optimization, RAG/LLM automation, cost-efficient scaling. Currently solving: High-throughput payment processing reliability (66.78% recovery via predictive analytics). Comfortable writing scalable APIs, designing for failure, and debugging production incidents.

TECHNICAL SKILLS

Languages	Python, C, JavaScript, SQL
AI/LLM	Retrieval-Augmented Generation (RAG), LLM Automation, Prompt Engineering, Embeddings, Tokenization, OpenAI, Anthropic
Frontend	React, HTML5, CSS3, Tailwind CSS
Backend	Flask, FastAPI, Node.js, Express, Prisma, REST APIs
Databases	PostgreSQL, MySQL, SQLite
Developer Tools	Git, GitHub, Docker, GitHub Actions, Linux, VS Code, Postman, npm, Vercel, Render
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, System Design

PROJECTS

VendorBridge – Scaled SaaS Platform from MVP → 50K Concurrent Users in 6 Months

Problem: Manual RFQ processing bottleneck preventing vendor scaling

Accomplished: Architected full-stack procurement platform handling 200K+ monthly transactions with sub-100ms API latency

Measured by:

- 99.7% uptime across production (6-month track record)
- 8-minute quotation turnaround (vs. 2-hour manual process)
- Successfully onboarded 50+ concurrent vendors without infrastructure degradation

Doing:

- Designed PostgreSQL schema with strategic indexing for high-throughput RFQ processing; load-tested to 500+ QPS
- Built stateless REST APIs (Node.js/Express) with rate limiting and circuit breakers; implemented distributed tracing for production debugging
- Architected approval workflows as finite state machines with immutable audit logs for compliance
- Deployment: CI/CD pipeline (GitHub Actions) with automated testing; blue-green deployment strategy on Render
- Architecture decision: PostgreSQL (transactional consistency) over MongoDB for payment workflows

RECLAIM – Intelligent Payment Failure Recovery System

Accomplished: Developed data-driven payment recovery system identifying high-impact recovery actions from failed transaction patterns

Measured by:

- Achieved 66.78% recovery rate on historical test dataset (10K+ transactions analyzed)
- Prioritized recovery opportunities reducing estimated customer churn by 15% through predictive intervention
- Processing 1,000+ transactions per minute with sub-second latency using vectorized operations

Doing:

- Built analytical pipelines (Python/Pandas) extracting customer, payment method, and transaction features; segmented risk cohorts using clustering
- Implemented supervised learning model with feature engineering for predicting recovery success; evaluated precision/recall trade-offs
- Deployed analysis on production transaction database with schema design for efficient historical queries and time-series windowing

ClearSky AI – AI-Powered Environmental Analysis Platform

Accomplished: Built AI-powered environmental analysis platform enabling image uploads and actionable sustainability insights with audit-ready report generation

Measured by:

- Processed 500+ image analyses in beta testing with 95%+ user satisfaction on insight accuracy
- Reduced manual environmental assessment time from 30 minutes to <2 minutes per image
- Generated 100+ downloadable PDF reports with zero formatting errors across all browsers

Doing:

- Designed REST API (FastAPI) with request validation, error handling, and graceful degradation for LLM rate limits
- Implemented modular backend architecture separating image ingestion, AI analysis, and report generation as independent microservices
- Built React frontend with image caching strategy and optimistic UI updates; deployed with Vercel CDN for sub-second load times
- Database: SQLite for MVP with migration path to PostgreSQL documented for production scaling

ToolGuard-AI – Industrial Tool Condition Monitoring System

Accomplished: Developed computer vision classification system predicting industrial cutting-tool conditions from images for predictive maintenance without sensor hardware

Measured by:

- Achieved 77.50% classification accuracy on held-out test set with stratified k-fold validation (0% false negatives for 'dulled' class)
- Reduced tool replacement costs by estimated 18% through early wear detection
- Processed 100+ tool images per inference run in batch mode at <50ms per image

Doing:

- Fine-tuned ResNet18 architecture using transfer learning and data augmentation (rotation, lighting variation) for real-world robustness
- Implemented class-balanced training with stratified cross-validation; created evaluation pipelines with per-class precision/recall metrics
- Built regression model for wear-measurement estimation with ablation studies comparing feature importance across CNN layers
- Deployment: Containerized model (Docker) for easy integration; designed inference API with fallback logic for ambiguous predictions

TECHNICAL LEADERSHIP & GROWTH

- **Google Student Ambassador (2026–2027)** — Led workshops on full-stack development and system design for 200+ students; mentored 5 junior developers on production deployment patterns
- **DSA Problem-Solver** — Solved 50+ LeetCode problems (focus: Dynamic Programming, Graph Algorithms); maintained 85%+ first-attempt success rate on medium/hard difficulty problems
- **Hackathon Track Record** — Smart India Hackathon 2026: Top 10 finalist for waste management optimization platform with 8-document strategic submission
- **Active Contributor** — Maintains production software projects using Git and GitHub; practices continuous deployment and code review disciplines

EDUCATION

Bachelor of Engineering, Information Technology

LDRP Institute of Technology & Research • GPA: 8.01/10 (Top 10%) • Expected Graduation: May 2028